UNIT - 2

Probability Distributions

Jacob Bernoulli
(6 Jan'1655 – 16 Aug'1705)

Jacob Bernoulli was born in Basel, Switzerland, is a mathematician. He contradicted to the desires of his parents, studied mathematics and astronomy and made tremendous achievement in later years. He traveled throughout Europe from 1676 to 1682, learning about the latest discoveries in mathematics and the sciences.

Jacob Bernoulli's first important contributions were a pamphlet on the parallels of logic and algebra published in 1685, after he worked on probability and geometry in 1687 . Jacob Bernoulli's paper of 1690 is important for the history of calculus, and in that integration was used with its complete meaning. In 1683 Bernoulli discovered the constant e by studying a question about compound interest which required him to find the value of the following expression $(1 + 1/n)^{1/n}$ (which is in fact e).

'Statistical thinking will one day be as necessary for efficient citizenship as the ability to read and write'. – H.G. Wells

TYPES OF RANDOM VARIABLES

A random variable is of two types.

1. Discrete Random Variable
2. Continuous Random Variable

James (Jacob) Bernoulli (1654 - 1705)

Binomial distribution was developed by **James Bernoulli** (1654 – 1705) in the year 1700 and was first published in 1713 eight years after his death.

Notations

- n - Number of Trials
- X - Random Variable (Discrete), represents numbe of Success, which follows Binomial Distribution.
- p - Probability of Success in a single trial
- q - Probability of Failure in a single trial

Notations

- $p + q = 1 \longrightarrow q = 1 - p$
- N - Number of times "n" trials are repeated (or) Total number of sets.

Bernoulli Trial

A Bernoulli trial is a random experiment in which there are only two possible outcomes namely

1. Success
2. Failure

$$\text{Sample Space } S = \{ \text{ Success, } \text{ Failure } \}$$

Each trial is

- Mutually exclusive outcomes,
- Exhaustive outcomes.

Definitions

Bernoulli Experiment

The experiment consists of n independent repeated Bernoulli trials.

BINOMIAL DISTIBUTION

The Random variable X that counts the number of success, in the n Bernoulli trials is said to follow a Binomial distribution with parameters n and p written as $B(n, p)$

The Probability mass function

$$P(X = x) = P(x) = nC_x \; p^x \; q^{n-x}$$

where $x = 0, 1, 2, 3, \cdots, n$ with $p + q = 1$.

Assumptions of the Binomial distribution.

1. It has 2 possible outcomes **Success, Failure**
2. Number of trials are finite
3. Trials are Independent
4. The probability of Success is constant in any trial.

1. **Mean** $\mu = np$
2. **Variance** npq
3. **Standard Deviation** $= \sqrt{npq}$

1. **Mean** $>$ **Variance**

Example 1

In a certain city, 20% of population is literate, assume that 200 investigators take sample 10 of a individual to see they are literate. How many investigators would you expect to report that three people or less are literate in the sample?

Solution :

$$\begin{aligned}
\text{Let } X &= \text{a d.r.v. denotes the } \textbf{Number of literates}. \\
p &= \textbf{Probability of getting a literate}. \\
&= \textbf{Percentage of getting a literate} = \frac{20}{100} = 0.2. \\
q &= 1 - 0.2 = 0.8. \\
n &= \text{Sample size of people} = 10. \\
N &= \text{Number of investigators} = 200.
\end{aligned}$$

$\therefore X \sim$ B.D. (n, p).
W.K.T., the P.M.F. of the Binomial Distribution is

$$\begin{aligned} P(X = x) &= nC_x p^x q^{n-x} \\ &= 10C_x (0.2)^x (0.8)^{10-x}, \ x = 0, 1, 2, \cdots, n = 10. \end{aligned}$$

Probability to get 3 or less literates:

$$\begin{aligned} \therefore P(X \leq 3) =& P(X = 0) + P(X = 1) + P(X = 2) + P(X = 3) \\ =& 10C_0 (0.2)^0 (0.8)^{10} + 10C_1 (0.2)^1 (0.8)^9 \\ & + 10C_2 (0.2)^2 (0.8)^8 + 10C_3 (0.2)^3 (0.8)^7 \\ =& 0.17 + 0.268 + 0.302 + 0.201 \\ =& 0.8789 \end{aligned}$$

$$\begin{aligned} \therefore \text{No. of investigators reporting 3 or less literates} &= N \times P(X \leq 3) \\ &= 200 \times 0.8789 \\ &= 175.6 \\ &= 176 \end{aligned}$$

Example 2

Find Binomial distribution and $P(X = 4)$, for (a) mean 4, variance 3

Solution : Given distribution is Binomial.
We know that the PMF of the Binomial distribution is

$$P(X = x) = nC_x p^x q^{n-x}, x = 0, 1, 2, \cdots, n$$
$$\text{Mean of B.D.} = np$$
$$\text{Variance of B.D.} = npq$$

Relation between Mean & Variance is of B.D is Mean $>$ Variance.

(a) $$\text{Mean} = np = 4$$
$$\text{Variance} = npq = 3$$
$$4q = 3 \quad \therefore q = \frac{3}{4}$$
$$p = 1 - q = 1 - \frac{3}{4} = \frac{1}{4}$$
$$(5) \Rightarrow np = 4 \therefore n = 16$$
$$\therefore P(X = x) = 16C_x \left(\frac{1}{4}\right)^x \left(\frac{3}{4}\right)^{16-x} = 16C_4 \left(\frac{1}{4}\right)^4 \left(\frac{3}{4}\right)^{12} = 0.22$$

Example 3

The mean of a binomial distribution is 20 and standard deviation is 4. Find the parameters of the distribution.

Solution : Given, Mean $= np = 20.$ (1)

$$\begin{aligned}
\text{Standard deviation} &= \sqrt{\operatorname{Var}(X)} \\
\sqrt{npq} &= 4 \\
\Rightarrow npq &= 16 \\
\Rightarrow (20)\, q &= 16 && [\because \text{ by } (1)] \\
\Rightarrow q &= 4/5 \\
\therefore p &= 1 - q = 1/5 \\
\therefore (1) \Rightarrow n\,(1/5) &= 20 \\
\Rightarrow n &= 100
\end{aligned}$$

$\therefore$ The parameters are $n = 100$ and $p = 1/5$

Example 4

Four coins are tossed simultaneously. What is probability of getting (i) 2 heads, (ii) atleast 2 heads, (iii) atmost 2 heads.

Solution:

Let $X =$ **A r.v. denotes number of heads from four coins tossed**.

$$p = \textbf{Probability of getting a head while tossing a coin}$$
$$= \frac{1}{2} = 0.5.$$
$$q = 1 - p = 1 - 0.5 = 0.5.$$
$$n = \text{Number of coins tossed} = 4.$$

$\therefore X \sim$ B.D. (n, p).
W.K.T., the P.M.F. of the Binomial Distribution is

$$P(X = x) = nC_x p^x q^{n-x}, x = 0, 1, \cdots n$$
$$P(X = x) = 4C_x \left(\frac{1}{2}\right)^x \left(\frac{1}{2}\right)^{4-x}, x = 0, 1, 2, 3, 4.$$

(i) $P(2 \text{ heads}) = P(X = 2) = 4C_2 \left(\frac{1}{2}\right)^2 \left(\frac{1}{2}\right)^2 = 0.375.$

(ii) $P(\text{atleast } 2 \text{ heads}) = P(X \geq 2)$

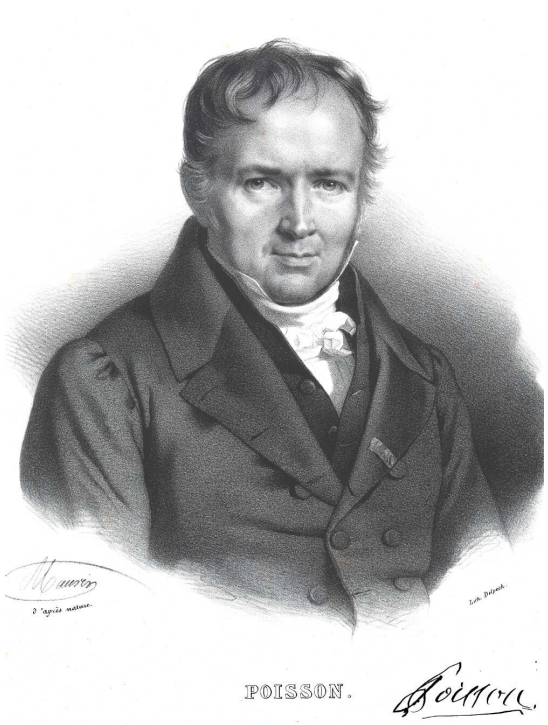
$$\begin{aligned}
&= 1 - P(X < 2) \\
&= 1 - [P(X = 0) + P(X = 1)] \\
&= 1 - \left[4C_0 \left(\frac{1}{2}\right)^0 \left(\frac{1}{2}\right)^4 + 4C_1 \left(\frac{1}{2}\right)^1 \left(\frac{1}{2}\right)^3\right] \\
&= 1 - [0.0625 + 0.25] = 1 - 0.3125 = 0.6875.
\end{aligned}$$

(iii) $P(\text{atmost } 2 \text{ heads}) = P(X \leq 2)$

$$\begin{aligned}
&= P(X = 0) + P(X = 1) + P(X = 2) \\
&= 4C_0 \left(\frac{1}{2}\right)^0 \left(\frac{1}{2}\right)^4 + 4C_1 \left(\frac{1}{2}\right)^1 \left(\frac{1}{2}\right)^3 + 4C_2 \left(\frac{1}{2}\right)^2 \left(\frac{1}{2}\right)^2 \\
&= 0.0625 + 0.25 + 0.375 \\
&= 0.6875.
\end{aligned}$$

Poisson Distribution

Siméon Denis Poisson (1781 - 1840)



Baron Siméon Denis Poisson; (21 June 1781 25 April 1840) was a French mathematician and physicist who worked on statistics, complex analysis, partial differential equations, the calculus of variations, analytical mechanics, electricity and magnetism, thermodynamics, elasticity, and fluid mechanics.

Conditions to use Poisson Distribution

1. The random variable X should be discrete.
2. The probability of the success (p) of the event is very very small.
3. The number of trials n, becomes very very large
4. Here np is a finite quantity and not necessarily large.

POISSON DISTRIBUTION

The probability distribution of a random variable X is said to have a Poisson distribution if it takes only non-negative values and if its distribution is given by

The Probability mass function

$$P(X = x) = \frac{e^{-\lambda} \cdot \lambda^x}{x!}, x = 0, 1, 2, \cdots$$

where $\lambda = np$ is the parameter and $e = 2.7183$

Statistical measures of the Poisson distribution.

1. **Mean** $\lambda = np$
2. **Variance** $\lambda = np$
3. **Standard Deviation** $\sqrt{np}$ **or** $\sqrt{\lambda}$

Some Examples of Piosson Distribution

1. The number of wrong telephone calls in a day.
2. The number of deaths in a city in one year by a rare disease.
3. The number of defective screws per box of 10000 screws.
4. The number of printing mistakes in each page of a book.
5. The emission of radioactive (alpha) particles.
6. The number of earth quakes in a given time interval.

Questions for Practice

Example 1

In a blade factory there is a chance of $\frac{1}{500}$ for any blade to be defective. The blades are in packets of 10. Use Poisson distribution to calculate the number of packets that containing

(a) no defective

(b) one defective

(c) two defectives respectively in a consignment of 10,000 packets.

Solution : Let X =**Number of defective blade**
p = **Probability of getting a defective blade**
$= \frac{1}{500} = 0.002 \longrightarrow 0$
which indicates to use Poisson distribution
n= Number of blades in a packet =10 .
np = $10\left(\frac{1}{500}\right) = 0.02(= \lambda)$
N = Number of Total Packets = 10000
W.K.T., the P.M.F. of Poisson distribution is

$$P(X = x) = \frac{e^{-\lambda}\lambda^x}{x!} = \frac{e^{-0.02}\,(0.02)^x}{x!},$$

Probability of getting X number of defectives	**Number of packets with X number of defectives**
(a) $P(X=0) = \dfrac{e^{(-0.02)}(-0.02)^0}{0!}$ $=0.98019.$	(a) $N \times P(X=0)$ $= 10000 \times 0.98019$ $= 9801.9$ $\cong 9802.$
(b) $P(X=1) = \dfrac{e^{-0.02}(-0.02)^1}{1!}$ $=0.0196.$	(b) $N \times P(X=1)$ $= 10000 \times 0.0196$ $= 196.$
(c) $P(X=2) = \dfrac{e^{(-0.02)}(-0.02)^2}{2!}$ $=0.00019603.$	(c) $N \times P(X=2)$ $= 10000 \times 0.00019603$ $= 1.9603$ $\cong 2.$

Example 2

In turning out certain toys in a manufacturing process in a factory, the average number of defectives is 10%. What is the probability of getting exactly 3 defective toys in a sample of 10 toys chosen at random by using Poisson distribution.

Solution: Let X be a d.r.v. denotes the number of defective toys. Given average number of defective toys is 10%.

$$\lambda = 10\% = \frac{10}{100} = 0.1 \qquad (\because \text{ Mean of the P.D. is } \lambda)$$

W.K.T., the P.M.F. of Poisson Distribution is

$$P(X = x) = \frac{e^{-\lambda}\lambda^x}{x!}, x = 0, 1, 2, \cdots \infty$$

$$\therefore P(X = 3) = \frac{e^{-(0.1)}(0.1)^3}{3!} = 1.508 \times 10^{-4} = 0.0001508.$$

Example 3

A manufacturer of pins knows that 2% of his products are defective. If he sells pins in boxes of 100 and guarantees that not more than 4 pins will be defective. What is the probability that the box will fail to meet the guarantee quality.

Solution: Let X be a d.r.v. **denotes the number of pins**.

$$p = \textbf{Probability of getting a defective pin} = 2\% = \frac{2}{100}$$

$$n = \text{number of pins} = 100$$

$$\lambda = np = \frac{2}{100} \times 100 = 2$$

$$\lambda = 2$$

The Manufacturer guarantees that not more than 4 defective pins in a box of 100.
Fail to meet the guarantee quality means that number of defectives are more than 4 in a box of 100.
We have to find the probability of number of defectives more than 4.

By Poisson's distribution, the P.M.F. is

$$P(X = x) = \frac{e^{-\lambda}\lambda^x}{x!}, x = 0, 1, 2, \cdots \infty$$

$\therefore P(X > 4) = P(X = 5, 6, \cdots, 100)$ which takes more time to calculate

$$\begin{aligned}
\therefore P(X > 4) &= 1 - P(X \leq 4) \qquad \text{(alternative formula)} \\
&= 1 - \begin{bmatrix} P(X = 0) + P(X = 1) + P(X = 2) \\ +P(X = 3) + P(X = 4) \end{bmatrix} \\
&= 1 - \left[\frac{e^{-2}2^0}{0!} + \frac{e^{-2}2^1}{1!} + \frac{e^{-2}2^2}{2!} + \frac{e^{-2}2^3}{3!} + \frac{e^{-2}2^4}{4!}\right] = 0.0527
\end{aligned}$$

$P(X > 4) = 0.0527.$

Example 4

X is a Poisson distribution random variable such that $P(X=1)=0.3, P(X=2)=0.2$. Find $P(X=0)$.

Solution : Given X is a Poisson distribution random variable.

$$\text{i.e., } X \sim \text{P.D.}(\lambda = np)$$

W.K.T., the probability mass function of Poisson distribution is

$$P(X=x) = \frac{e^{-\lambda}\lambda^x}{x!}, x = 0, 1, 2, 3, \cdots, \infty$$

Given $P(X=1) = 0.3$ (1)

$P(X=2) = 0.2$ (2)

For that

$(1) \Rightarrow P(X=1) = 0.3$ $\frac{e^{-\lambda}\lambda^1}{1!} = 0.3$ $e^{-\lambda}\lambda = 0.3 \quad \cdots (3)$	$(2) \Rightarrow P(X=2) = 0.2$ $\frac{e^{-\lambda}\lambda^2}{2!} = 0.2$ $e^{-\lambda}\lambda^2 = 0.4 \quad \cdots (4)$

$$\frac{(4)}{(3)} \Rightarrow \frac{e^{-\lambda}\lambda^2}{e^{-\lambda}\lambda} = \frac{0.4}{0.3} = \frac{4}{3}$$

$$\lambda = \frac{4}{3}$$

We have to find $P(X=0) = \frac{e^{-\lambda}\lambda^0}{0!} = e^{-\lambda} = e^{-4/3} = 0.263.$

Problem No. 1

Problem No. 1

Problem 1 Fit a binomial distribution for the following data:

x :	0	1	2	3	4	5	6
f :	5	18	28	12	7	6	4

Solution:

The binomial distribution is defined as

$$P(X = x) = nC_x p^x q^{n-x}, x = 0, 1, 2..., n.$$

Find the mean of binomial distribution.

$$\mu = \overline{x} = E(x) = \sum x.p(x)$$

$$Mean = \overline{x} = \frac{\sum fx}{\sum f} = \frac{\sum fx}{N}$$

The binomial distribution,

$$P(X = x) = nC_x p^x q^{n-x}, x = 0, 1, 2..., n.$$
$$= 6C_x (0.4)^x (0.6)^{6-x}, x = 0, 1, 2..., 6.$$

Expected frequency: $N * P(X = x) = 80 * 6C_x (0.4)^x (0.6)^{6-x}$

Problem No. 1

x	f	fx	Theorecital Frequencies	
0	5	0	3.732	**4**
1	18	18	14.9296	**15**
2	28	56	24.8832	**25**
3	12	36	22.1184	**22**
4	7	28	11.0592	**22**
5	6	30	2.9488	**3**
6	4	24	0.328	**0**
	$\sum f = 80$	$\sum fx = 192$		

SRM INSTITUTE OF SCIENCE & TECHNOLOGY (Deemed to be University u/s 3 of UGC Act, 1956)

Problem No. 2

Problem 2 Fit a Poisson distribution for the following data:

Deaths:	0	1	2	3	4	5
Frequency:	142	156	69	27	5	1

Solution:

The Poisson distribution is defined as

$$P(X = x) = \frac{e^{-\lambda}\lambda^x}{x!}, \quad x = 0, 1, 2, \ldots$$

Find the mean of Poisson distribution.

$$\mu = \overline{x} = E(x) = \sum x.p(x)$$

$$Mean = \overline{x} = \frac{\sum fx}{\sum f} = \frac{\sum fx}{N}$$

$$N = 142 + 156 + 69 + 27 + 5 + 1 = 400$$

$$\sum fx = 0 + 156 + 138 + 81 + 20 + 5 = 400$$

$$\lambda = \overline{x} = \frac{400}{400} = 1$$

The Poisson distribution,

$$P(X = x) = \frac{e^{-\lambda}\lambda^x}{x!}, \quad x = 0, 1, 2, \ldots$$
$$= \frac{e^{-1}(1)^x}{x!}, \quad x = 0, 1, 2, \ldots, 5.$$

Expected frequency: $N * P(X = x) = 400 * \frac{e^{-1}(1)^x}{x!}$

Problem No. 1

x	f	fx	Theorecital Frequencies	
0	142	0	147.1518	147
1	156	156	147.1518	147
2	69	138	73.57589	74
3	27	81	24.5253	25
4	5	20	6.1313	6
5	1	5	1.2263	1
	$\sum f = 400$	$\sum fx = 400$		

SRM
INSTITUTE OF SCIENCE & TECHNOLOGY
(Deemed to be University u/s 3 of UGC Act, 1956)

Exponential Distribution

A continuous random variable X is said follow exponential distribution with parameter $\lambda \geq 0$, if its p.d.f. is given by

$$f(x) = \begin{cases} \lambda e^{-\lambda x}, & x \geq 0, \lambda > 0 \\ 0, & \text{otherwise} \end{cases}$$

Note : Symbolically, $X \sim$ Exponential distribution (λ). Exponential distribution is also known as negative exponential distribution.

Example 1

Problem

If X be the time to repair (in hours) with p.d.f. $f(x) = ce^{-x/5}$, $x > 0$, find: (a) c

(b) Mean (c) Variance (d) $P(X > 3)$ (e) $P(3 < X \leq 6)$

Solution:

Solution

$$\text{Given } f(x) = ce^{-x/5},\ x > 0$$

$$\int_0^\infty f(x)\,dx = 1$$

$$c\int_0^\infty e^{-x/5}dx = 1$$

$$c\left[\frac{e^{-x/5}}{-\frac{1}{5}}\right]_0^\infty = 1$$

$$5c = 1$$

$$\therefore\ c = \frac{1}{5}$$

Solution

$$f(x) = \frac{1}{5}e^{-x/5} = \lambda e^{-\lambda x}, \quad \lambda = \frac{1}{5}$$

$$\therefore X \sim \text{Exponential}\left(\frac{1}{5}\right)$$

$$\text{Mean} = \frac{1}{\lambda} = 5$$

$$\text{Variance} = \frac{1}{\lambda^2} = 25$$

Solution

$$\text{(d)} \quad P(X > 3) = \int_3^\infty \frac{1}{5} e^{-x/5} dx = \frac{1}{5}\left[\frac{e^{-x/5}}{-\frac{1}{5}}\right]_3^\infty = -\left[e^{-x/5}\right]_3^\infty$$

$$= -\left(0 - e^{-3/5}\right) = e^{-3/5} = e^{-0.6} = 0.5488$$

$$\text{(e)} \quad P(3 < X \leq 6) = \int_3^6 \frac{1}{5} e^{-x/5} dx$$

$$= e^{-3/5} - e^{-6/5}$$

$$= e^{-0.6} - e^{-1.2}$$

$$= 0.2476$$

Memory Less Property of Exponential distribution

State and prove Memory Less Property of Exponential distribution

Statement : If X is an exponentially distributed Random Variable then

$$P(X > s + t/X > s) = P(X > t)$$

for any $s, t > 0$.

Proof : W.K.T., the pdf of Exponential distribution is

$$f(x) = \lambda e^{-\lambda x}, x > 0$$

Memory Less Property of Exponential distribution

$$R.H.S. : P(X > t)$$

$$P(X > t) = \int_t^\infty f(x)dx$$

$$= \int_t^\infty \lambda e^{-\lambda x} dx = \lambda \left[\frac{e^{-\lambda x}}{-\lambda} \right]_t^\infty$$

$$= -\left[0 - e^{-\lambda(t)} \right]$$

$$= e^{-\lambda(t)}$$

Memory Less Property of Exponential distribution

Similarly, we can say

$$P(X > s + t) = e^{-\lambda(s+t)}$$

$$P(X > s) = e^{-\lambda s}$$

Memory Less Property of Exponential distribution

$$\textbf{L.H.S. :}\ P(X > s + t/X > s)$$

$$= \frac{P(X > s + t \cap X > s)}{P(X > s)}$$

$$= \frac{P(X > s + t)}{P(X > s)}$$

$$= \frac{e^{-\lambda(s+t)}}{e^{-\lambda s}} = \frac{e^{-\lambda s}e^{-\lambda t}}{e^{-\lambda s}} = e^{-\lambda t}$$

$$P(X > s + t/X > s) = P(X > t)$$

Note: If X is exponential distributed random variable, then

$$P(X > t) = e^{-\lambda t}, \lambda > 0$$

Example 2

Example 2

The mileage of a tyre have exponential distribution with mean 40,000 kms. Find the probability that one of these tyres will last

(a) atleast 20,000 kms (b) atmost 30,000 kms

(c) atleast 30,000 kms., given that it has been used for 20,000 kms.

Statistical measures of the Normal distribution.

Solution : Given distribution is exponential.
Let X be a random variable which denotes the mileage of a tyre.
W.K.T., the pdf of exponential distribution with parameter λ is

$$f(x) = \lambda e^{-\lambda x}, x \geq 0.$$

$$\text{Given Mean } = 40000 \text{ kms}$$

$$\text{i.e., } \frac{1}{\lambda} = 40000$$

$$\lambda = \frac{1}{40000}$$

$$\therefore f(x) = \frac{1}{40000} e^{-\frac{x}{40000}}$$

$$\text{(a) } P(X \geq 20000) = e^{-\frac{20000}{40000}}$$

$$= e^{-\frac{1}{2}} = 0.6065$$

$$\text{(b) } P(X \leq 30000) = 1 - P(X > 30000)$$

$$= 1 - e^{-\frac{30000}{40000}} = 1 - e^{-\frac{3}{4}} = 0.5276$$

$$\text{(c) } P(X > 30000/X > 20000)$$

$$= P(X > 20000 + 10000/X > 20000)$$

$$= P(X > 10000) = e^{-\frac{10000}{40000}} = e^{-\frac{1}{4}}$$

$$= 0.7788$$

Gaussian Distribution (or) Normal Distribution

Normal distribution is also known as **Gaussian distribution** named after Carl Friedrich Gauss. The normal distribution as we know it today was first described by Carl Friedrich Gauss, a German mathematician, physicist, and astronomer. In his work "**Theoria Motus Corporum Coelestium**" published in **1809**, Gauss introduced the normal distribution as a probability distribution for errors in astronomical observations. He derived the formula for the probability density function of the normal distribution and demonstrated its properties.

Conditions

1. The random variable X should be continuous.
2. n, the number of trials is infinitely large, i.e., $n \to \infty$
3. Neither p(or q) is very small, i.e., p and q are fairly near equal.

NORMAL DISTRIBUTION

A continuous random variable X with parameters μ and σ^2 is said to follow a normal distribution if its probability density function is given by

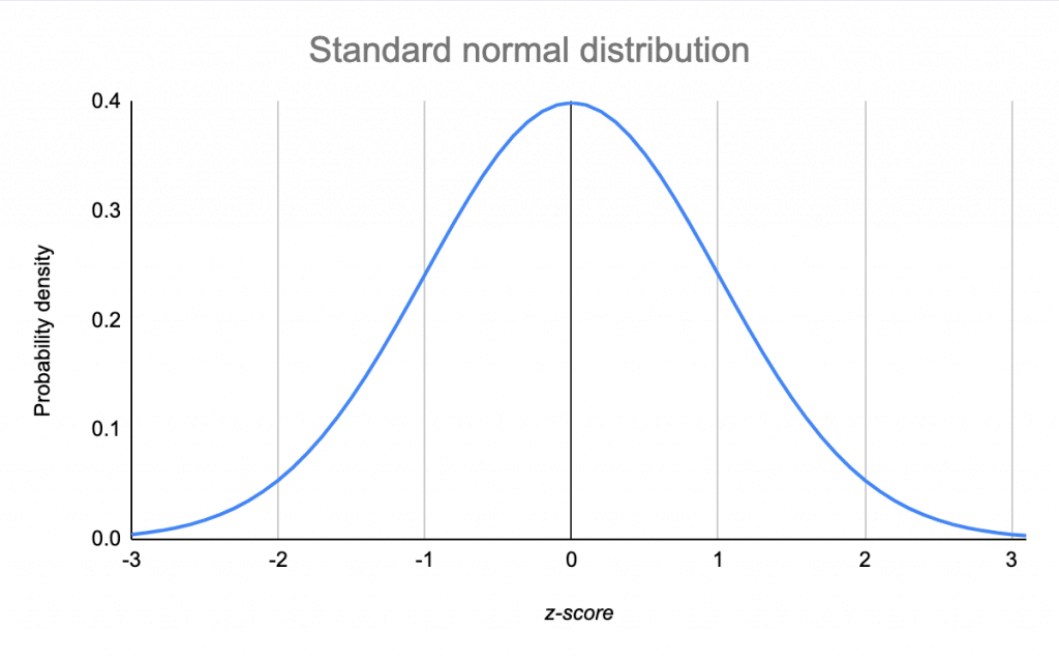
$$f(x) = \frac{1}{\sigma\sqrt{2\pi}}\, e^{-\frac{1}{2}\left(\frac{x-\mu}{\sigma}\right)^2}$$

where $-\infty < x < \infty$ and $\sigma > 0$

Also it is represented as $X \sim N(\mu, \sigma^2)$

Statistical measures of the Normal distribution.

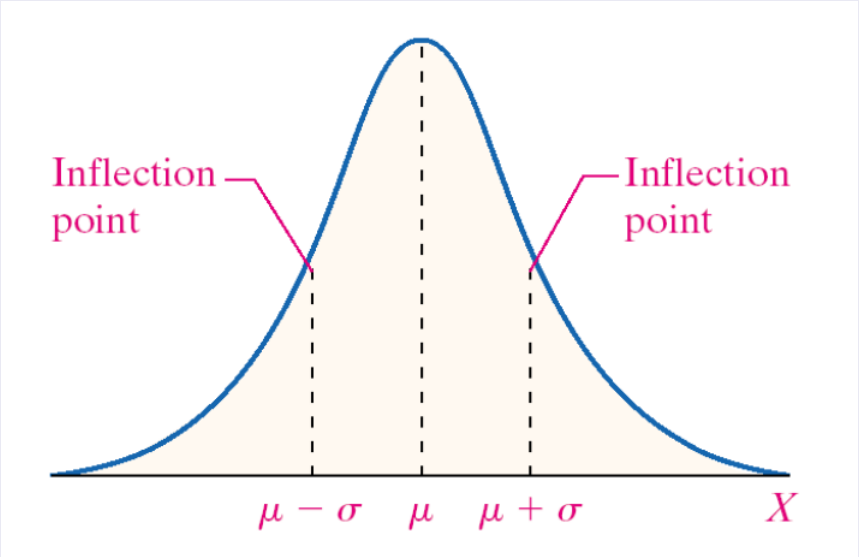
1. It is a bell shaped curve



2. It is symmetric about $x = \mu$

Statistical measures of the Normal distribution.

1. The points of inflexion of the curve are at $x = \mu - \sigma$ and $x = \mu + \sigma$



2. The curve approaches the horizontal axis asymptotically on either side of $x = \mu$

Statistical measures of the Normal distribution.

1. Area under the curve is one
2. The mean, median and mode coincide at $x = \mu$
3. The curve of normal distribution has a single peak, i.e., it is a unimodal.
4. The curve approaches the horizontal axis asymptotically on either side of $x = \mu$
5. The maximum value of $f(x)$ is at $x = \mu$ and the maximum value is

$$\frac{1}{\sigma\sqrt{2\pi}}$$

STANDARD NORMAL VARIATE (Z)

Any random variable whose

1. Mean $\mu = 0$.
2. standard deviation $\sigma = 1$.

is called standard random variable

It is represented as $X \sim N(0, 1)$

Conversion Formula

$$Z = \frac{X - \mu}{\sigma}$$

Standard Normal Distribution

$\therefore X \sim N(0,1)$ is called Standard normal Random Variable.

Definition :

If $X \sim N(\mu, \sigma^2)$ then $\boxed{Z = \dfrac{X-\mu}{\sigma}}$ is called standard normal random variable and its probability density function is

$$\phi(z) = \frac{1}{\sqrt{2\pi}} e^{\frac{-z^2}{2}}, \; -\infty < z < \infty$$

The standard normal distribution is also known as **Unit Normal Distribution** or **Z - Distribution.**

Example 1

Refer area under Normal curve for $P(0 < Z < z_1)$ at the end of this chapter. Find

(a) $P(0 < Z < 1.25)$ (b) $P(0.6 < Z < 1.25)$

(c) $P(Z < 1.25)$ (d) $P(Z > 1.25)$

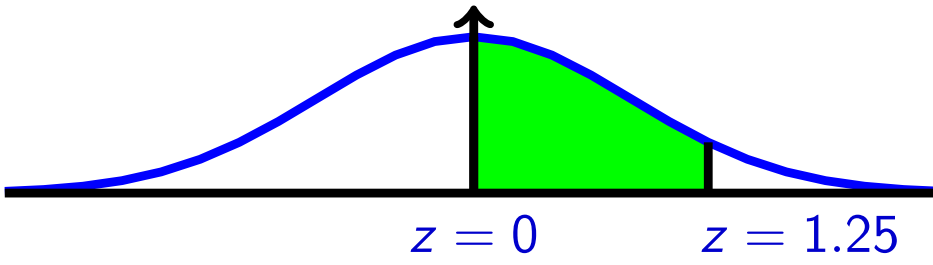
(e) $P(|Z| < 1.25)$ (f) $P(-2.5 < Z < 1.25)$

(g) $P(z = 1.25)$

Solution : Here Z is continuous normal variable.

(a) $P(0 < Z < 1.25)$

$= 0.3944$

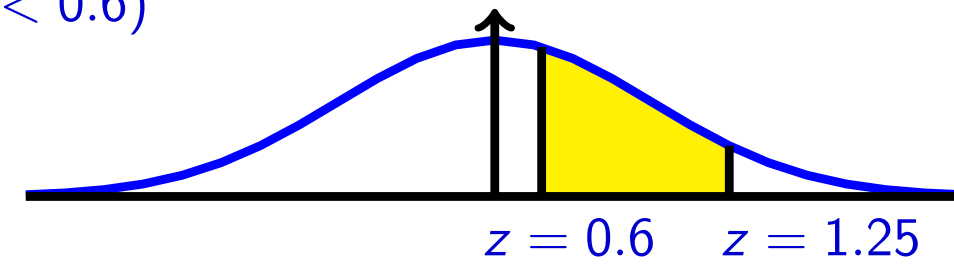


(b) $P(0.6 < Z < 1.25)$

$= P(0 < Z < 1.25) - P(0 < Z < 0.6)$

$= 0.3944 - 0.2258$

$= 0.1686$

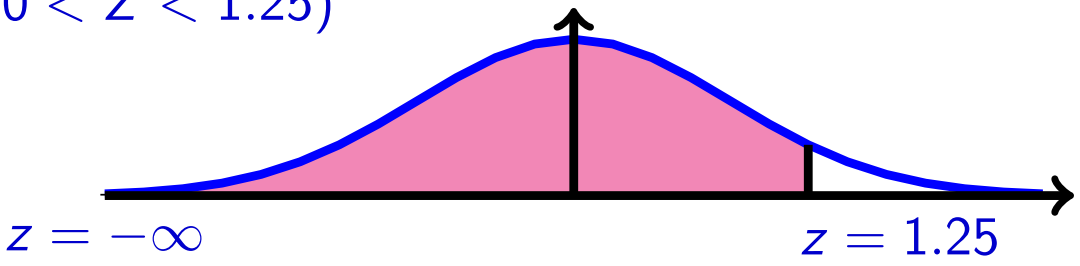


(c) $P(-a < Z < a)$

$= P(-\infty < Z < 0) + P(0 < Z < 1.25)$

$= 0.5 + 0.3944$

$= 0.8944$

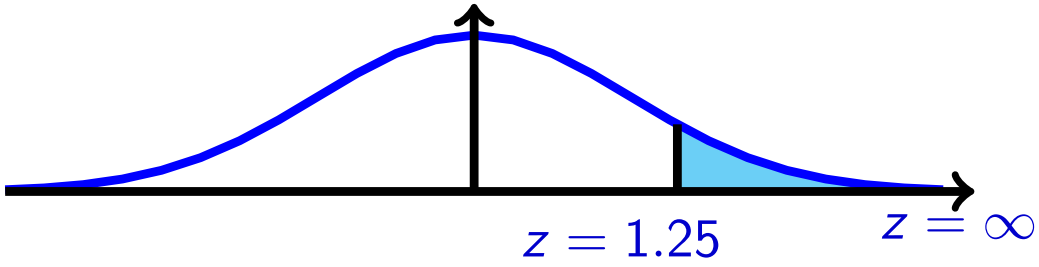


(d) $P(Z > 1.25)$

$= 1 - P(Z < 1.25)$

$= 1 - 0.8944 \qquad [\because \text{ by } (c)]$

$= 0.1.56$



(e) $P(|Z| < 1.25)$

$= P(-1.25 < Z < 1.25)$

$= P(-1.25 < Z < 0) + P(0 < Z < 1.25)$

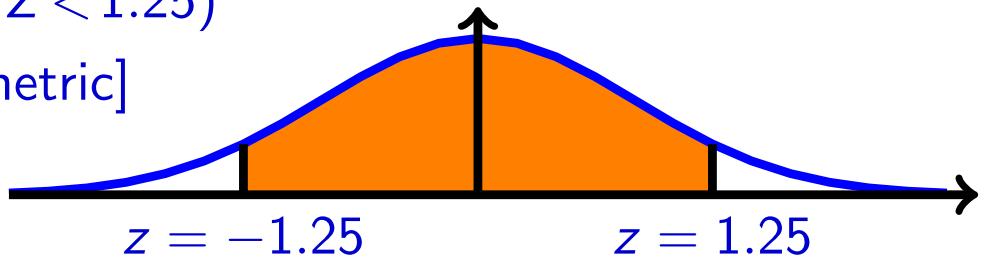
$= P(0 < Z < 1.25) + P(0 < Z < 1.25)$

[$\because$ Normal curve is symmetric]

$= 2 \times P(0 < Z < 1.25)$

$= 2 \times 0.3944$

$= 0.7888$



(f) $P(-2.5 < Z < 1.25)$

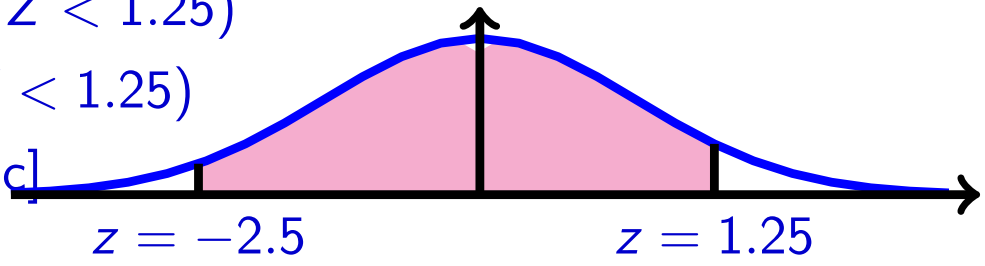
$= P(-2.5 < Z < 0) + P(0 < Z < 1.25)$

$= P(0 < Z < 2.5) + P(0 < Z < 1.25)$

[$\because$ Normal curve is symmetric]

$= 0.4938 + 0.3944$

$= 0.8882$



(g) $P(Z = 1.25) = 0$ [$\because P(Z = \text{constant}) = 0$]

Example 2

The life of a certain kind of electronic device has a mean of 300 hours and standard deviation of 25 hours. Assuming thta the life time of the devices follow normal distribution.

1. Find the probability that any one of these devices will have a life time more than 350 hours.
2. What is the percentage will have life time between 220 and 260 hours?
3. What is the percentage will have life time between 325 and 350 hours?

Solution : Given $\mu = 30, \sigma = 5 \therefore X \sim \mathrm{N}(\mu, \sigma^2)$
Let

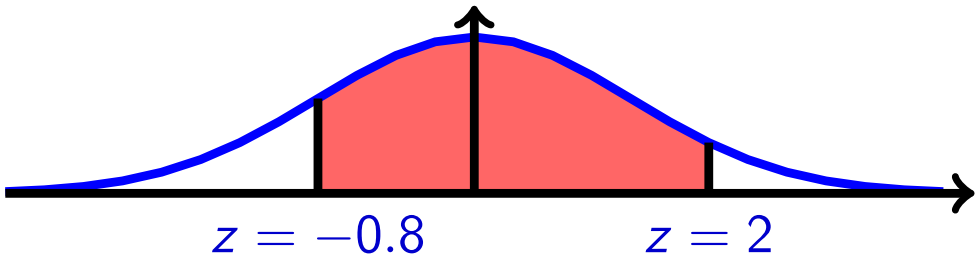
$$\boxed{Z = \frac{X - \mu}{\sigma} = \frac{X - 30}{5}}$$

(a) $P(26 < X < 40) = P\left(\frac{-4}{5} < Z < \frac{10}{5}\right)$

$$= P(-0.8 < Z < 2)$$
$$= P(-0.8 < Z < 0) + P(0 < Z < 2)$$
$$= P(0 < Z < 0.8) + P(0 < Z < 2)$$

[$\because$ Normal curve is symmetric]

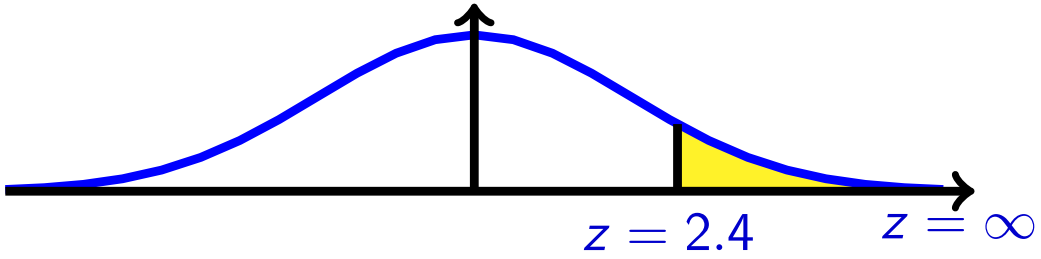
$$= 0.2881 + 0.4773$$
$$= 0.7654$$



(b) $P(X > 45)$

$$= P(Z > 3) = 0.5 - P(0 < Z < 3)$$

$$= 0.0013$$



(c) $P(|X-30|>5) = 1 - P(|X-30|<5)$

$[\because$ by modulus definition$]$

$= 1 - P(-5 < X - 30 < 5)$

$= 1 - P(25 < X < 35)$

$= 1 - P(-1 < Z < 1)$

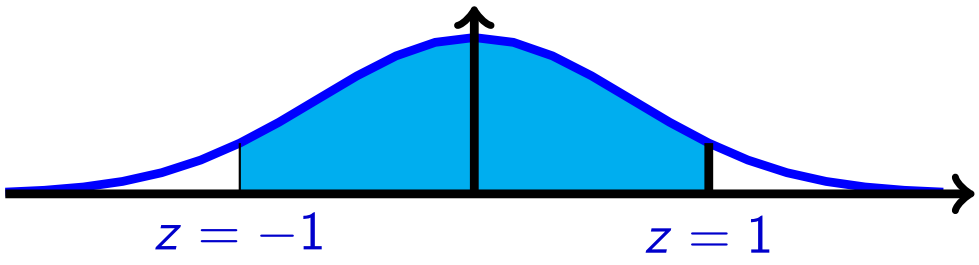
$= 1 - 2 \times P(0 < Z < 1)$

$[\because$ Normal curve is symmetric$]$

$= 1 - 2 \times 0.3431$

$= 1 - 0.6862$

$= 0.3138$



Example 3

In a normal distribution, 31% of items are under 45 and 8% are over 64. Find mean and variance.

Solution :

Given distribution is Normal.

Given that 31% of items are under 45 and 8% are over 64.

We have to find μ, σ^2.

W.K.T., $\boxed{Z = \frac{X - \mu}{\sigma}}$

When $X_1 = 45$	When $X_2 = 64$
Let $Z = -z_1 = \dfrac{X_1 - \mu}{\sigma}$ $[\because (1)]$ $= \dfrac{45 - \mu}{\sigma}$ (2) Given, $P(Z < -z_1) = 31\% = 0.31$ i.e., $P(-z_1 < Z < 0) = 19\% = 0.19$ i.e., $P(0 < Z < z_1) = 19\% = 0.19$ By table, $\therefore z_1 = 0.5, -z_1 = -0.5,$ $\dfrac{45 - \mu}{\sigma} = -0.5$ $[\because (2)]$ $45 - \mu = -0.5\sigma$ $\mu - 0.5\sigma = 45$ (3)	Let $Z = z_2 = \dfrac{X_2 - \mu}{\sigma}$ $[\because (1)]$ $= \dfrac{64 - \mu}{\sigma}$ (4) i.e., $P(Z > z_2) = 8\% = 0.08$ i.e., $P(0 < Z < z_2) = 42\% = 0.42$ By table, $\therefore z_2 = 1.4$ $\dfrac{64 - \mu}{\sigma} = 1.4$ $[\because (4)]$ $64 - \mu = 1.4\sigma$ $\mu + 1.4\sigma = 64$ (5)

Solving (3) and (5), Mean $= \mu = 50$

S.D. $= \sigma = 10$

Variance $= \sigma^2 = 100$

Example 4

In a normal distribution, 7% of items are under 35 and 89% are under 63. Find μ, σ^2.

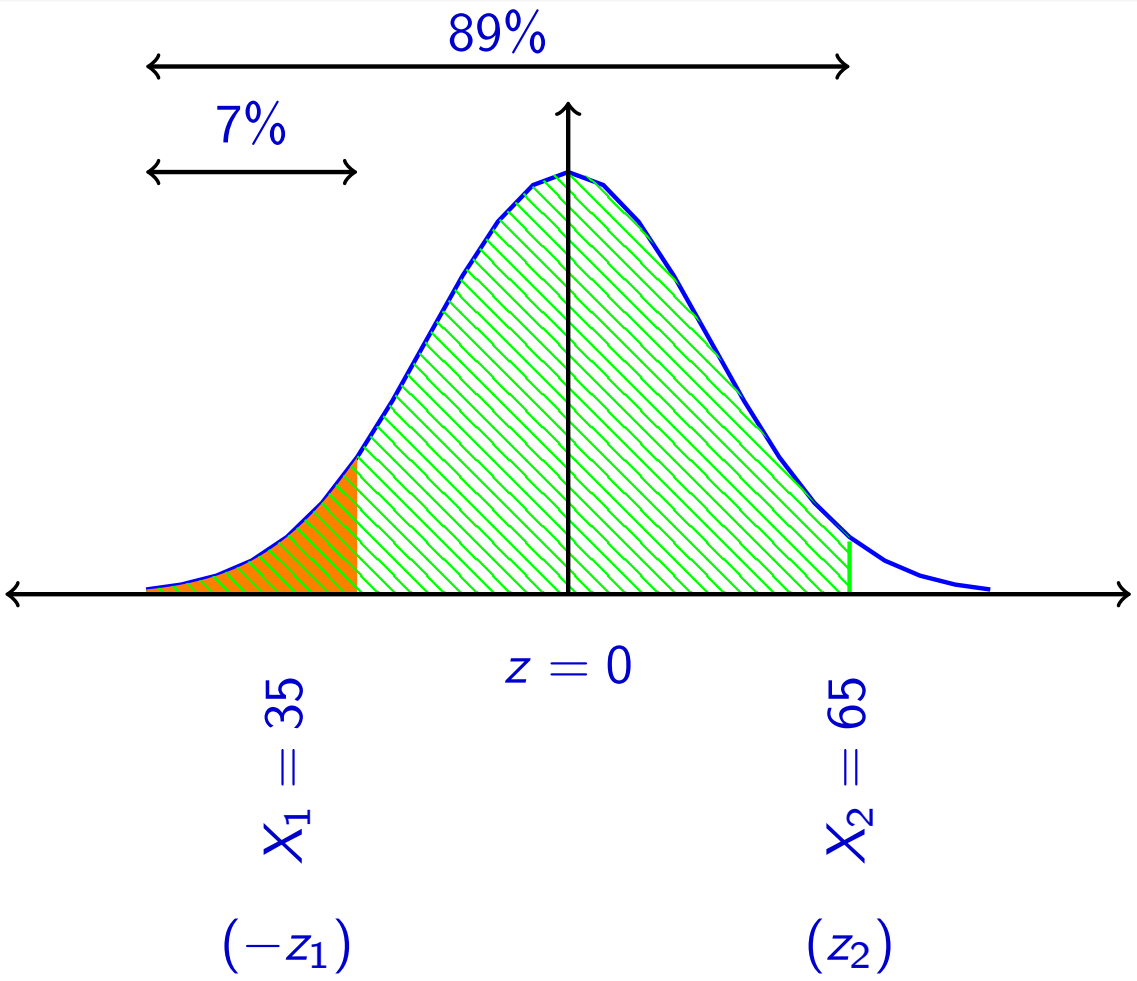
Solution :

Given distribution is Normal.

Given that 7% of items are under 35 and 89% are under 63.

We have to find μ, σ^2.

W.K.T., $\boxed{Z = \frac{X - \mu}{\sigma}}$


89%
7%
$z = 0$
$X_1 = 35$
$(-z_1)$
$X_2 = 65$
(z_2)

When $X_1 = 35$	When $X_2 = 65$
$\text{Let } Z = -z_1 = \dfrac{X_1 - \mu}{\sigma} \quad [\because (1)]$ $= \dfrac{35 - \mu}{\sigma} \quad (2)$ $\text{i.e., } P(Z < -z_1) = 7\% = 0.07$ $\text{i.e., } P(-z_1 < Z < 0) = 43\% = 0.43$ $\text{i.e., } P(0 < Z < z_1) = 43\% = 0.43$ From table, $\therefore z_1 = 1.48,$ $-z_1 = -1.48$ $\dfrac{35 - \mu}{\sigma} = -1.48 \quad [\because (2)]$ $35 - \mu = -1.48\sigma$ $\mu - 1.48\sigma = 35 \quad (3)$	$\text{Let } Z = z_2 = \dfrac{X_2 - \mu}{\sigma} \quad [\because (1)]$ $= \dfrac{65 - \mu}{\sigma}$ $\text{i.e., } P(Z < z_2) = 89\% = 0.89$ $\text{i.e., } P(0 < Z < z_2) = 39\% = 0.39$ From table, $\therefore z_2 = 1.23$ $\dfrac{65 - \mu}{\sigma} = 1.23 \quad [\because (4)]$ $65 - \mu = 1.23\sigma$ $\mu + 1.23\sigma = 65$

Solving (3) and (5), Mean $= \quad \mu = 51.3837$

S.D. $= \quad \sigma = 11.0701$

Variance $= \quad \sigma^2 = 122.5474$